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In [8]: from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Load dataset
data = load_diabetes()

X = data.data
y = data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create model
model = LinearRegression()

# Train model
model.fit(X_train, y_train)

# Predict
predictions = model.predict(X_test)

# Accuracy
score = r2_score(y_test, predictions)

print("Model Accuracy:", score)
```

Model Accuracy: 0.4526027629719195

Predictive Analysis using Machine Learning

This project demonstrates a machine learning model using Linear Regression to predict outcomes based on a dataset. The model is trained using Scikit-learn and evaluated using R^2 score

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In [9]: import matplotlib.pyplot as plt

plt.scatter(y_test, predictions)

plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted")

plt.show()
```

